

SMART HOSPITAL SYSTEM

Course: CSC 1012 - Introduction to Computer Programming

GitHub URL: <https://github.com/nimethpiyumal/CSC1012-Smart-Hospital-System>

1. Project Overview

For this assignment, I built a C program called "Smart Hospital & Resource Allocation System". The main idea of this system is to help hospital staff register new patients, assign them beds in different wards, sort them based on emergency priority, and calculate their final bill. I designed the code using modular programming without using any C structs or dynamic memory.

2. System Structure

I split the code into several smaller `.c` files to keep it clean and easy to manage. All shared variables and functions are connected using one header file (`hospital.h`):

- `hospital.h`: Holds global constants, shared variables, and function declarations.
- `hospital.c`: Sets up initial data like doctor fees, daily bed costs, and empty beds.
- `patient.c`: Takes user input to register patients and checks if age inputs are valid.
- `billing.c`: Calculates the doctor fee, extra charges for emergency cases, ward costs, and age discounts.
- `sorting.c`: Sorts the patient queue based on how urgent their health condition is.
- `reports.c`: Shows hospital statistics like total money earned and bed usage.
- `main.c`: Shows the interactive menu for the user to select options.

3. Data Storage & Logic

- **Parallel Arrays:** Since I did not use C `struct` types, I used multiple parallel arrays with the same array index `i` to store patient details (for example: `patientName[i]`, `patientAge[i]`, `urgencyLevel[i]`, and `assignedWard[i]`).
- **Bed Occupancy Matrix:** I used a 2D array `bedOccupancy[4][20]` to keep track of 80 total beds across 4 wards. An array value of 0 means the bed is free, and 1 means it is taken.
- **Priority Queue Sorting:** I chose Bubble Sort to arrange patients by urgency level (Level 3 = Critical, Level 2 = Urgent, Level 1 = Normal). When two patients switch places in the queue, I swap their data across all parallel arrays at the same time to keep records accurate.
- **Billing Logic:**
 - Urgent patients (Level 2) get a 20% surcharge, while Critical patients (Level 3) get a 50% surcharge on doctor fees.
 - Patients under 5 years old or over 65 years old receive a 15% discount on their total bill.

4. Limitations & Assumptions

- The system can store up to 100 patients at a time (`MAXPATIENTS = 100`).
- Data is stored in temporary memory (RAM), so all patient records reset when the program is closed.
- Emergency (Level 3) patients get immediate help, so their estimated wait time is set to 0 minutes.

5. How to Compile and Run

❖ Main menu

```
=====
      SMART HOSPITAL RESOURCE ALLOCATION SYSTEM
=====
1. Register New Patient
2. View Bed Occupancy Status
3. Print Patient Bill & Invoice
4. Sort Patient Queue by Priority (Urgency Level)
5. View Summary Reports & Analytics
6. Exit
=====
Enter your choice (1-6): |
```

❖ Bed Status

```
===== HOSPITAL BED OCCUPANCY STATUS =====

General Ward (Capacity: 20):
[Bed 01: FREE] [Bed 02: FREE] [Bed 03: FREE] [Bed 04: FREE] [Bed 05: FREE]
[Bed 06: FREE] [Bed 07: FREE] [Bed 08: FREE] [Bed 09: FREE] [Bed 10: FREE]
[Bed 11: FREE] [Bed 12: FREE] [Bed 13: FREE] [Bed 14: FREE] [Bed 15: FREE]
[Bed 16: FREE] [Bed 17: FREE] [Bed 18: FREE] [Bed 19: FREE] [Bed 20: FREE]

Paediatric Ward (Capacity: 10):
[Bed 01: FREE] [Bed 02: FREE] [Bed 03: FREE] [Bed 04: FREE] [Bed 05: FREE]
[Bed 06: FREE] [Bed 07: FREE] [Bed 08: FREE] [Bed 09: FREE] [Bed 10: FREE]

Surgical Ward (Capacity: 10):
[Bed 01: FREE] [Bed 02: FREE] [Bed 03: FREE] [Bed 04: FREE] [Bed 05: FREE]
[Bed 06: FREE] [Bed 07: FREE] [Bed 08: FREE] [Bed 09: FREE] [Bed 10: FREE]

ICU (Intensive Care Unit) (Capacity: 5):
[Bed 01: FREE] [Bed 02: FREE] [Bed 03: FREE] [Bed 04: FREE] [Bed 05: FREE]
=====
```

❖ Final Bill

```
=====
                        SMART HOSPITAL ADMISSION & BILL
=====
Patient ID           : PAT-1001
Patient Name        : kamal
Age                 : 22 Years
Specialty           : Cardiology
Assigned Ward       : Paediatric Ward (Bed #01)
Urgency Level       : Level 2 (Urgent)
=====
Base Consultation Fee : LKR      4500.00
Emergency Surcharge   : LKR      900.00
Ward Stay Cost (2 Days): LKR     12000.00
=====
Gross Total Bill      : LKR     17400.00
Age Subsidy Discount  : LKR -      0.00
=====
Final Payable Amount  : LKR     17400.00
Estimated Waiting Time : 0.00 mins
=====
```

❖ Summary Report

--- HOSPITAL SUMMARY REPORT ---	
Total Registered Patients	: 3
- Level 1 (Normal)	: 1
- Level 2 (Urgent)	: 0
- Level 3 (Critical)	: 2
Financial Summary:	
- Total Revenue Earned	: LKR 87500.00
- Total Discounts Given	: LKR 0.00
Ward Occupancy Rates:	
- General Ward:	0.0% (0/20 beds)
- Paediatric Ward:	20.0% (2/10 beds)
- Surgical Ward:	10.0% (1/10 beds)
- ICU (Intensive Care Unit):	0.0% (0/5 beds)
Highest Paying Patient	: saman (LKR 53000.00)